

10	D	As the sphere moves from highest to lowest point, increase in K.E. = decrease in G.P.E. $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mg(2L)$ $v^2 = 2\left(2gL + \frac{1}{2}u^2\right)$ When the sphere is at the lowest point, $T - mg = \frac{mv^2}{L}$ $T = \frac{mv^2}{L} + mg$ $= \frac{2m\left(2gL + \frac{1}{2}u^2\right)}{L} + mg$ $= \frac{2(0.40)\left(2(9.81)(0.50) + 0.5(2.5)^2\right)}{0.50} + (0.40)(9.81)$ $= 24.62 = 25 \text{ N}$
11	D	GMm